

Recitation 1

RL Basics and Markov Decision Processes

ESE 3990 · Fall 2026

Plan for the recitation

- 1 RL basics: loop, rewards, state, the Markov property
- 2 Markov chains and Markov reward processes
- 3 Value functions and the Bellman equation
- 4 MDPs, policies, and the Bellman optimality
- 5 Quiz – 30 minutes
- 6 Quiz answer keys

Part 1 · What's a reinforcement learning problem

- Sequential.
 - Evaluative feedback.
 - Your action changes the data.
-

Which of these are RL problems?

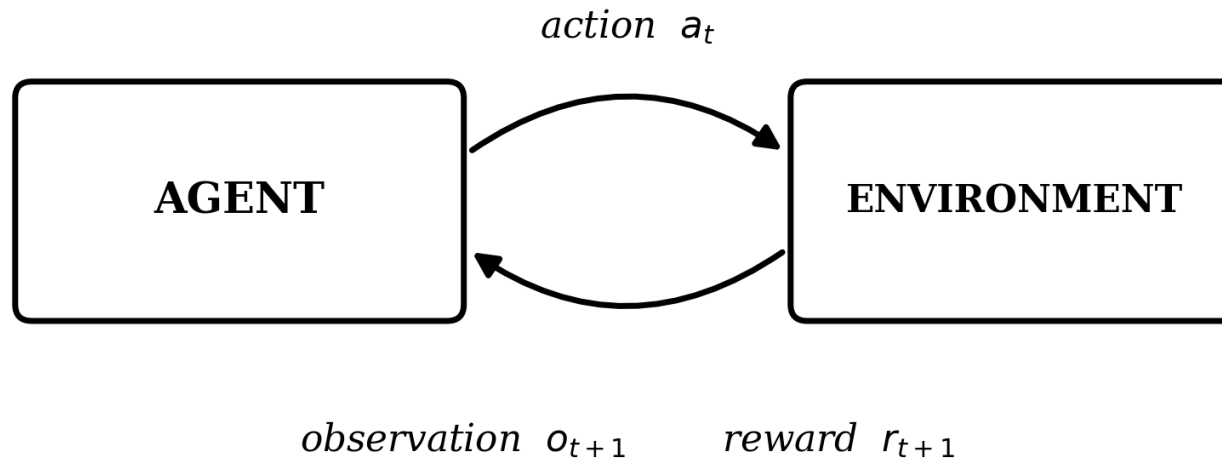
A robot learns to flip a pancake by trying, dropping it, and trying again.

YES

A model is trained on 10,000 labeled X-rays to classify new ones.

no — supervised

The agent–environment loop



"What should I do?"

"Here is what happened."

Index convention. The agent acts at time t , so the next observation and the reward it triggers are labeled $t+1$.

The goal. Choose actions to maximize **cumulative future reward**.

History, state, and the Markov property

$$h_t = (o_0, a_0, r_1, o_1, a_1, r_2, \dots, a_{t-1}, r_t, o_t)$$

A state is any function of the history: $s_t = f(h_t)$.

Environment state	s_t^e	The environment's own private bookkeeping.
Agent state	s_t^a	The agent's internal summary.

The Markov property

$$P[s_{t+1} | s_t] = P[s_{t+1} | s_1, s_2, \dots, s_t]$$

Check. If I handed you this state and erased your memory of everything before it, could you predict the future just as well?

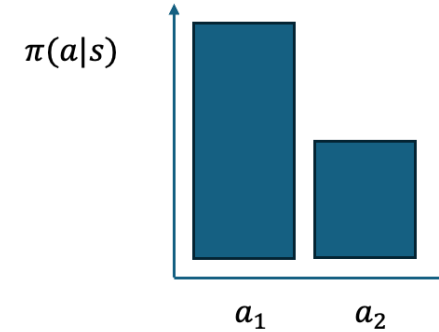
Index notation: $r_{(t+1)}$ convention — a_t produces r_{t+1} and o_{t+1} , so the newest reward inside h_t is r_t .

What is inside an agent

Policy

$$\pi(a | s)$$

"How I behave."



Value function

$$V(s) \text{ or } Q(s, a)$$

"How good is this?" A prediction of total future reward.

Model

$$p(s_{t+1} | s_t, a_t)$$

"What happens if I...?" The agent's prediction of the environment.

Value functions

$$V(s) = \mathbb{E}[r_{t+1} + r_{t+2} + \dots \mid s_t = s]$$

"How much total reward from here?"

If you only know V you cannot act optimally without a model.

$$Q(s, a) = \mathbb{E}[r_{t+1} + r_{t+2} + \dots \mid s_t = s, a_t = a]$$

"How much if my first action is a ?"

If you know Q you can act optimally with no model at all — just compare $Q(s, a)$ across the available actions.

Part 2 · Building the MDP

Markov process (Markov chain)	$\langle S, P \rangle$	A random walk between states.
Markov reward process (MRP)	$\langle S, P, R, \gamma \rangle$	Add a reward for each state.
Markov decision process (MDP)	$\langle S, A, P, R, \gamma \rangle$	Add actions. Now you choose, and your choice changes both the reward and where you go next.
Partially observable MDP (POMDP)	$\langle S, A, O, P, R, Z, \gamma \rangle$	Add observations. You no longer see the state and must infer it.

The return: what we are maximizing

$$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$

G is a number for ONE episode.

- it is a random variable.
- the first reward is never discounted

What the discount factor does

$\gamma = 0$ Completely myopic.

$\gamma \approx 0.9$ The agent effectively looks about ten steps ahead.

$\gamma = 1$ Perfectly far-sighted. Only safe if every episode is guaranteed to end.

Effective planning horizon: $T = 1 / (1 - \gamma)$.

Compute every return in an episode

An episode lasts five steps and then ends, so $T = 5$. The rewards received are:

t	1	2	3	4	5
r_t	-1	+2	+6	+3	+2

The discount factor is $\gamma = 0.5$.

Find every return in the episode: $G_0, G_1, G_2, G_3, G_4, G_5$

Hint — try forwards and backwards, which one saves computes.

Compute every return in an episode

$$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots = r_{t+1} + \gamma (r_{t+2} + \gamma r_{t+3} + \dots) = r_{t+1} + \gamma G_{t+1}$$

So every return is the next reward plus the discounted return that follows it.

G_5	$= 0$	the episode has ended	0
G_4	$= r_5 + 0.5 G_5$	$= 2 + 0.5(0)$	2
G_3	$= r_4 + 0.5 G_4$	$= 3 + 0.5(2)$	4
G_2	$= r_3 + 0.5 G_3$	$= 6 + 0.5(4)$	8
G_1	$= r_2 + 0.5 G_2$	$= 2 + 0.5(8)$	6
G_0	$= r_1 + 0.5 G_1$	$= -1 + 0.5(6)$	2

The rewards add up to 12, but G_0 is only 2 — the big rewards arrive late.

With $\gamma = 1$ the answer would be 12; with $\gamma = 0$ it would be -1.

Keep this recursion in mind — it is the Bellman equation, applied to a single episode.

Part 3 · Value functions and the Bellman equation

$$V(s) = \mathbb{E}[G_t \mid s_t = s]$$

the definition of value

$$= \mathbb{E}[r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots \mid s_t = s]$$

write out the return

$$= \mathbb{E}[r_{t+1} + \gamma (r_{t+2} + \gamma r_{t+3} + \dots) \mid s_t = s]$$

factor γ out of everything after the first term

$$= \mathbb{E}[r_{t+1} + \gamma G_{t+1} \mid s_t = s]$$

the bracket IS the return starting at t+1

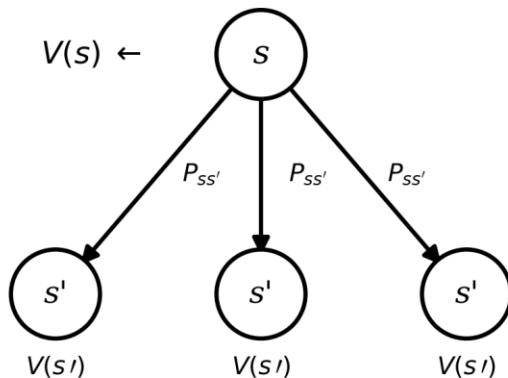
$$= \mathbb{E}[r_{t+1} + \gamma V(s_{t+1}) \mid s_t = s]$$

and its expectation is the value of the next state

$$= R(s) + \gamma \sum_{s'} P_{ss'} V(s')$$

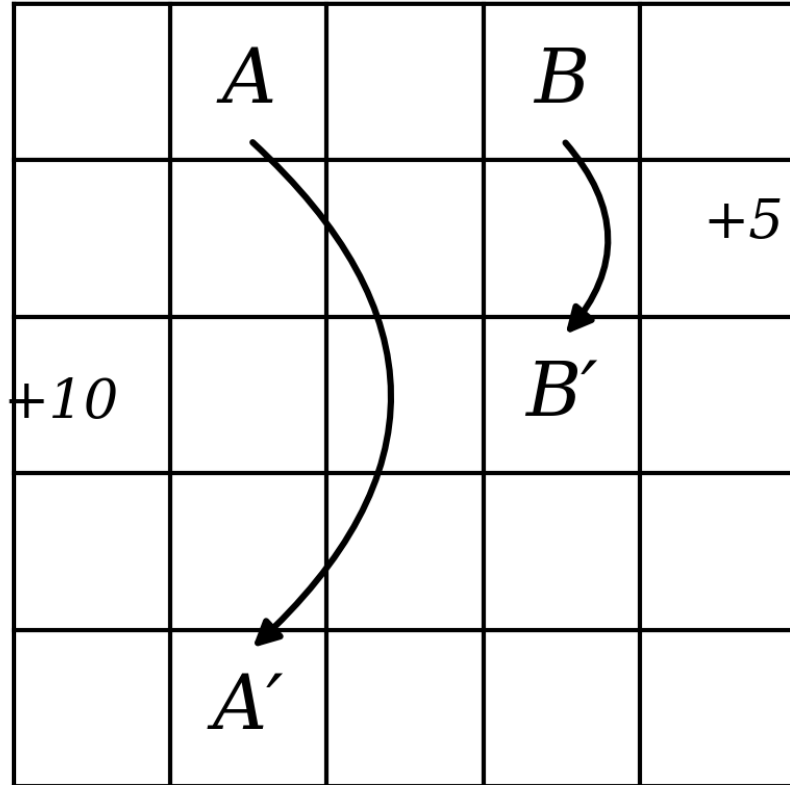
where $R(s) = \mathbb{E}[r_{t+1} \mid s_t = s]$ is the expected reward collected on leaving s .

reward r collected here



- The value of being here = what I collect now + (discounted) the value of wherever I land next.
- The equation is recursive.

Grid World: an example



- Four actions in every square: N, S, E, W.
- A move that would take you off the grid leaves you where you are and pays -1 .
- From A , every action pays $+10$ and jumps you to A' . From B , every action pays $+5$ and jumps you to B' .
- Every other move pays 0 .
- The discounting factor $\gamma = 0.9$.

Verifying the value of the center square

Gridworld under the uniform random policy, $\gamma = 0.9$.

3.3	8.8	4.4	5.3	1.5
1.5	3.0	2.3	1.9	0.5
0.1	0.7	0.7	0.4	-0.4
-1.0	-0.4	-0.4	-0.6	-1.2
-1.9	-1.3	-1.2	-1.4	-2.0

The policy picks each direction with probability 0.25

$$V(s) = \sum_a 0.25 [r(s,a) + \gamma V(s')]$$

From the center every move is legal, so every reward is 0

$r = 0$ for all four actions

The four neighbors are 2.3 (N), 0.4 (E), -0.4 (S), 0.7 (W)

$$V(s) = 0.25 \times 0.9 \times [2.3 + 0.4 + (-0.4) + 0.7]$$

Arithmetic

$$V(s) = 0.25 \times 0.9 \times 3.0 = 0.675 \approx 0.7$$

Policies

$$\pi(a | s) = P[a_t = a | s_t = s]$$

- It may be deterministic or stochastic.
- It depends only on the current state, not on the history.
- It is stationary: the same rule at every time step.

Fix a policy, and an MDP becomes an MRP

$$\langle S, A, P, R, \gamma \rangle + \text{a policy } \pi = \langle S, P^\pi, R^\pi, \gamma \rangle$$

$$P_{ss'}^\pi = \sum_a \pi(a|s) P_{ss'}^a$$

the chance of going $s \rightarrow s'$, averaged over which action π would pick

$$R_s^\pi = \sum_a \pi(a|s) R_s^a$$

the reward you expect from s , averaged the same way

Value functions and Bellman equations

$$V^\pi(s) = \mathbb{E}_\pi[G_t \mid s_t = s]$$

Start in s and then behave according to π forever after.

$$Q^\pi(s, a) = \mathbb{E}_\pi[G_t \mid s_t = s, a_t = a]$$

Start in s , take the action a , and follow π ever after.

$$V^\pi(s) = \sum_a \pi(a|s) Q^\pi(s, a)$$

The Bellman expectation equation

$$V^\pi(s) = \sum_a \pi(a|s) [r(s, a) + \gamma \sum_{s'} P_{ss'}^a V^\pi(s')]$$

Note there is no max anywhere. This equation evaluates the policy you were handed. It does not try to improve it.

Recommend: derive from the definition.

Optimality: the best you could possibly do

$$V^*(s) = \max_{\pi} V^{\pi}(s)$$

the most total reward obtainable from s under the best behavior

$$Q^*(s, a) = \max_{\pi} Q^{\pi}(s, a)$$

the most obtainable if you must take a first, then behave optimally

- There always exists an optimal policy π^*
- All optimal policies achieve V^* and Q^* .
- There is always a deterministic optimal policy.

$$\pi^*(a|s) = 1 \text{ if } a = \operatorname{argmax}_a Q^*(s, a), \quad \text{otherwise } 0$$

- If you know Q^* , you already have the optimal policy.
- With only V^* you cannot act.
- Solving an MDP means finding V^* or Q^* .

Expectation vs optimality

BELLMAN EXPECTATION

$$V^\pi(s) = \sum_a \pi(a|s) [r + \gamma \sum_{s'} P V^\pi(s')]$$

BELLMAN OPTIMALITY

$$V^*(s) = \max_a [r + \gamma \sum_{s'} P V^*(s')]$$

What question does it answer?

How good is THIS policy?

How good is the BEST policy?

Over actions, it...

AVERAGES, weighted by π

takes the MAXIMUM

Do you need a policy first?

Yes

No — it finds one

Is it linear?

Yes

No — max is not a linear operator

Closed-form solution?

Yes — $V^\pi = (I - \gamma P^\pi)^{-1} R^\pi$

None in general

One word — sum or max — is the entire difference.

Important concepts

Reward vs. return

r is one step; G is the whole sum from now to the end.

V vs. Q

V takes a state; Q takes a state AND an action.

Sum vs. max

Expectation **averages** over actions using π . Optimality takes the **max**.

The missing superscript

V^π is one policy's value; V^* is the best among all policies.

The first reward is undiscounted

The exponent starts at 0, not 1.

$\gamma = 1$ is not always safe

Only if every episode is guaranteed to terminate.

"Markov" is about the STATE

It is a property of the representation you chose, not of the environment.

Feedback

Google form link:

